BEING A CLOSED IMMERSION IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one Morphisms of Spaces lemma and one Descent lemma plus same-chapter items, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Framework lemma for descending properties of morphisms. Stacks Project, Descent on Spaces, Lemma lemma-descending-properties-morphisms. Supplies the criterion structure that the proof verifies.
- Closed immersions are \'etale local on the base. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-closed-immersion-local. Supplies assumptions (1) and (2).
- fpqc descent of affineness. Stacks Project, Descent on Spaces, Lemma lemma-descending-property-affine. Makes X a scheme with $X \to Z$ affine.
- fpqc descent of closed immersions for schemes. Stacks Project, Descent, Lemma descent-lemma-descending-property-closed-immersion. Concludes $X \to Z$ is a closed immersion.

Source reference. Original lemma 041J: sources/stacks-chow/tags.txt, source lines 5294–5294. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03M4: sources/stacks-chow/tags.txt, source lines 4736–4736. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 041Z: sources/stacks-chow/tags.txt, source lines 5310–5310. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02L6: sources/stacks-chow/tags.txt, source lines 3397–3397. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is a closed immersion" is fpqc local on the base.*

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Proof. We will use Lemma 041J to prove this. Assumptions (1) and (2) of that lemma follow from Morphisms of Spaces, Lemma 03M4. Consider a cartesian diagram

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow & & \downarrow \\ Z' & \longrightarrow & Z \end{array}$$

of algebraic spaces over S where $Z' \to Z$ is a surjective flat morphism of affine schemes, and $X' \to Z'$ is a closed immersion. We have to show that $X \to Z$ is a closed immersion. The morphism $X' \to Z'$ is affine. Hence by Lemma 041Z we see that X is a scheme and $X \to Z$ is affine. It follows from Descent, Lemma 02L6 that $X \to Z$ is a closed immersion as desired. □